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RESEARCH ARTICLE

Simulating human responses to environmental messaging

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Abstract
This paper presents ongoing work to implement and evaluate virtual humans whose responses to environmental messaging are shaped by their media diets and social interactions. The project scraped thousands of social media post-comment pairs related to environmental issues, classified them by viewpoint through the large-scale orchestration of multiple instances of large language models, and built a vector database of embedded interactions with associated classification metadata to serve as a knowledge source for a chatbot. Dynamic, metadata-based filtering of this knowledge source, in conjunction with retrieval-augmented generation, enabled a chatbot with selectable personas that generate responses to new social media posts based on stereotypical viewpoints grounded in current news, attitudes and zeitgeists. A qualitative and quantitative evaluation was conducted to demonstrate the validity of the approach, though its full potential remains to be explored.

Keywords Climate · AI chatbots · Political discourse simulation · Computational social science

1 Introduction

Understanding human responses to environmental messaging is increasingly important as climate change discourse becomes more polarized. The effectiveness of communication hinges on message framing and resonance with different population segments. Social media platforms serve as valuable data sources for understanding public perceptions and trends related to environmental issues [1]. Public opinion

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on climate change varies considerably, and awareness doesn't always translate into strong policy support due to political polarization or perceived economic trade-offs [2, 3]. In this polarized landscape, understanding how different ideological groups respond to environmental messaging is crucial for effective communication strategies. Artificial Intelligence (AI) techniques, particularly machine learning and NLP [4], have been used to predict and analyse audience reactions to environmental messages. AI-powered chatbots and virtual assistants have engaged users in climate change education [5–9]. Social media platforms have recently seen the emergence of computer-generated avatars and virtual influencers utilising AI and machine learning [10, 11].

Large Language Models (LLMs) have recently emerged as a disruptive technology with remarkable abilities to understand complex linguistic structures and generate coherent responses [12]. However, despite being strong generalists, LLMs may use outdated knowledge and hallucinate (i.e. make things up) without specific external information. Retrieval-Augmented Generation (RAG) addresses this by integrating new more up to date and context specific information into LLM prompts via vector databases and similarity searches [13, 14]. Another emerging tool is AI agents which are autonomous software entities that utilize auxiliary LLMs for tasks such as data preparation [15].

The work presented in this paper leverages LLMs for classification and RAG to simulate public responses to environmental messaging on Reddit. The knowledge-base for our RAG chatbot is obtained by scraping Reddit post/comment pairs and using multiple instances of an LLM to classify each comment by viewpoint. The {posts/comment} pairs are embedded for RAG, and viewpoint classifications are stored as associated metadata, to be used as positive filters when accessing the vector database. In other words, the chatbot's knowledge base can be effectively sub sectioned by viewpoint to emulate archetypical users of social media.

As an initial proof of concept, we developed a chatbot capable of toggling between contrasting ideological priors on climate change (left/neutral/right), to emulate cross-ideological engagement. However, we acknowledge that climate worldviews are not necessarily reducible to a left/right binary. Movements such as the Gilets Jaunes in France highlight concerns about distributive justice in climate policy [16], while the American Conservation Coalition exemplifies "green conservatism" in the U.S. context [17].

Attitudes toward climate change exist on a broad spectrum of belief, concern, and engagement. For instance, early research in the U.S. grouped people into five interpretive communities based on their risk perception of various hazards, including global warming: the *Liberal Left*, *Alarmists*, *Mainstream Americans*, *Optimists*, and the *Religious Right* [18]. A more detailed analysis of beliefs, risks, policy preferences, and behaviors led to the now widely recognized Global Warming's Six Americas framework, which includes the *Alarmed*, *Concerned*, *Cautious*, *Disengaged*, *Doubtful*, and *Dismissive* segments [19]. The Six Americas framework has been consistently tracked over more than ten years, producing a strong set of long-term data and becoming a useful tool for both research and practice in climate communication [20]. Beyond the U.S., segmentation studies have identified parallel structures in other national contexts. For example [21], applied clustering to data from the UK House-

hold Longitudinal Study to identify three distinct clusters of climate risk attitudes: the *Concerned* (supportive of action), the *Paradoxical* (acknowledging risks but resistant to mitigation, reflecting ambivalence and low efficacy), and the *Sceptical* (minimising or dismissing risks).

Together, these frameworks illustrate both the stability of key audience segments (concerned vs. sceptical) and the value of intermediate categories that capture ambivalence. Such typologies provide a theoretical and empirical basis for simulating ideological engagement with climate messaging.

For this paper, 5000 randomly sampled Reddit post–comment pairs related to search queries about climate change were categorized. Categories and associated rubrics were based on a UK survey framework, with the categories *Concerned*, *Paradoxical*, and *Sceptical.* An additional category, *Irrelevant*, was specified to capture comments that did not pertain to the topic or were nonsensical. This dataset was used as the knowledge base for a retrieval-augmented generation (RAG) chatbot that allows users to select between these viewpoint archetypes.

This paper presents the methodology, an analysis of the validity of a large language model for classification, and an evaluation of the RAG chatbot's inferred alignment with human responses to Reddit posts. The software is configurable, allowing new searches, frameworks, categories, and rubrics to be easily specified.

2 Methods

2.1 Constructing the knowledge base

Reddit was our primary data source, its 430 million active users with 130,000 daily posts and 2.5 million comments offer a rich subset of ideologically polarized discussions on environmental issues. Using the Python Reddit API Wrapper (PRAW), we scraped post/comment pairs from subreddits focused on environmental topics like climate change, sustainability, and conservation. Once the data was collected, it underwent a processing pipeline consisting of filtering, classification and database construction (Fig. 1).

Firstly, irrelevant or unusable data was filtered out by excluding posts or comments containing URLs, external references, special characters and content of excessive length.

Classification was a batch process where an iterative cycle of LLMs calls was made to construct the dataset. We used the Python package LangChain to automate this process of LLM querying and hence classification. For development purposes we used Ollama to run instances of small resource LLMs (e.g. Magistral 24:B, and llama3.1:8B) on a high-performance computer with access to four Nvidia A100 powered GPUs. However, we later integrated support for API calls to popular large parameter cloud based LLMs (e.g. Gemini 2.5 Flash) which offered improved performance with classification.

Instructions and criteria for classifications were given, as a rubric, in a JSON file. This was hence parsed and incorporated into a prompt to the LLM instances responsible for classification. This approach gives the flexibility to specify new rubrics with

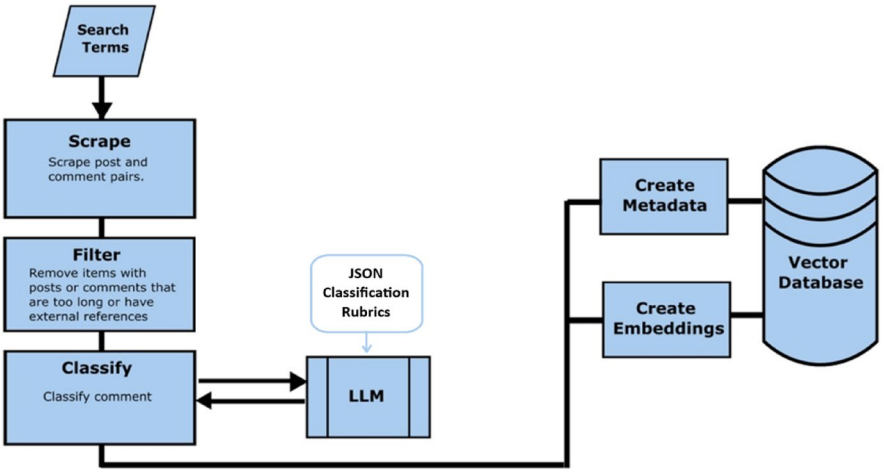


Fig. 1 Graph of the pipeline to construct the knowledge base of a RAG chatbot

Table 1 Rubric for reddit comment classification

Category	Description
Concerned	The comment acknowledges that climate change is real, imminent, and not exaggerated
	Expresses that urgent action is needed to address risks
	Strongly agrees that people will be affected within 30 years and that the crisis is not beyond control
Paradoxical	The comment holds mixed or ambivalent views
	Sometimes agrees with sceptical statements (e.g., crisis exaggerated), but also believes impacts are coming soon
	Expresses powerlessness, believing risks are real but out of human control
Sceptical	The comment downplays the risks of climate change
	Suggests impacts are exaggerated, distant, or not urgent
	Expresses low support for strong action.
	Corresponds to "Doubtful" or "Denial" groups in earlier research
Irrelevant	No relevant content to classify
	Nonsensical or too off-topic

multiple frameworks, for example uk_survey, us_5_communities, us_6_americas, etc. Table 1 shows the classification rubric for the uk_survey framework used in this paper.

The LLM prompt was built programmatically to include the Reddit **{post}**, associated **{comment}** to categorise with brief justifications, the **{rubric}** and instructions

to return JSON only format with an example of the JSON output. The expected JSON output can be easily constructed programmatically from the rubric's structure. The following shows the prompt template constructed for classifying comments.

```
The comment between the COMMENT markup tags is a reply to the post
between the POST markup tags.

<POST>{post}</POST> <COMMENT>{comment}</COMMENT>

Classify or score the COMMENT text (using the POST only as context). Use
the rubric between the RUBRIC markup tags to guide you. You are a JSON-
only assistant. Return ONLY a JSON object that matches this EXACT
structure: {JSON}

<RUBRIC>{rubric}</RUBRIC>
```

The following is an example of the JSON output from a classifying LLM. The JSON returned from the LLM was validated against a schema, so that rare malformed outputs were rejected.

```
[
    "categories": {
      "uk_3_cluster": "Sceptical",
    },
    "justifications": {
      "uk_3_cluster": "The comment expresses doubt about the
reliability of climate data, suggesting manipulation and comparing
climate science to conspiracy theories like the moon landing,
indicating skepticism."
    }
 ]
```

Early in the development we found that smaller locally run large language models were particularly prone to misclassify comments when sarcasm, humour or political tone inverted the literal meaning of the text. Also, these models struggled to identify the target of a comment and sometimes relied on surface lexical cues such as *"Republicans," "woke,"* or *"libs,"* to formulate a viewpoint without interpreting the overall meaning a given comment. In some cases, the models also hallucinated detailed justifications consistent with their chosen label rather than drawing on evidence from the comment itself. To mitigate this, we added an 'extra_instructions' field to the JSON rubric and hence the LLMs prompts, For example, directing the LLM to look for the target of strong negative, sarcastic, or aggressive comments and not to automatically assume a sceptical viewpoint from dismissive or cynical comments. In the results section we assess categorisation when building the knowledge base by comparing samples with human classifications.

It should be noted at this stage that although there are misclassified items, the knowledge source will later be used for retrieval augmented generation where several candidate items will be taken from the knowledge source, and the prevailing view will be inferred in generating a given comment. Hence, mitigating for the influence of individual erroneous classifications.

The categorisations for each post/comment pair were added to its associated metadata (ids, timestamps, etc.), and hence the resultant record was added to a ChromaDB vector database. The associated post and comment text was embedded, i.e. converted to numerical representation based on semantic meaning, and so positioned into the high-dimensional vector space of the vector database. We chose nomic-embed-text, a high-performing open embedding model with a large token context window.

2.2 The RAG chatbot with personas

We developed the chatbot to use a local large language model (for example gpt-oss:20b) and the LangChain framework for RAG. The chatbot was deployed using the Streamlit web API providing a user-friendly interface. The user selects a classification (Concerned, Paradoxical or Skeptical) from a drop-down box and enters a hypothetical Reddit post into the chatbot.

When the user's new post is sent, the content of the vector database is positively filtered based on the classification. Hence the post is embedded to become amenable to a Maximal Marginal Relevance (MMR) search of the vector database. The MMR technique ensures that the selected items are not only relevant but also diverse, reducing redundancy in the results. This returns a list of $\boldsymbol{K}$ semantically matching real Reddit posts/comment items which are hence be added to the prompt which goes to the chatbot's LLM. Figure 2 shows how the prompt to the chatbot is constructed.

The following Prompt template was used, where {context} is the list of real Reddit posts/comment items from the database, which is injected into the prompt, and {question} is the original user's post given to the chatbot.

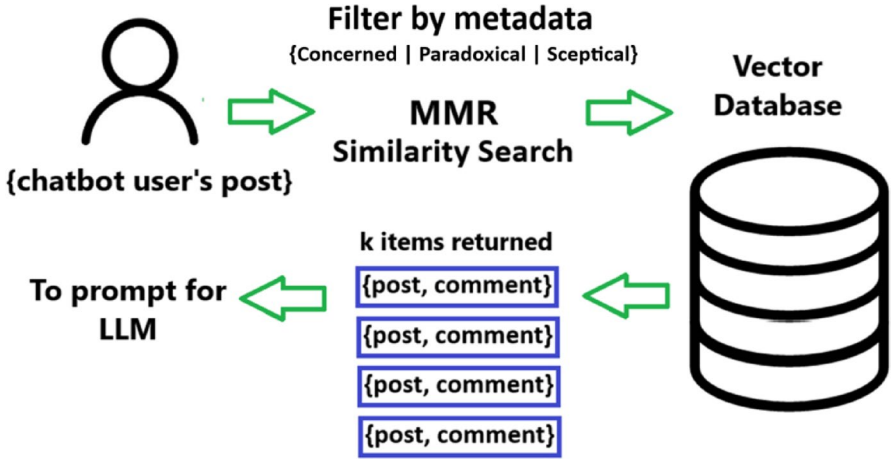


Fig. 2 A search filters based on metadata and uses MMR to find the most relevant post/comment items relative to the user's post. The returned items are the context to be added to the prompt for the LLM.

```
[SYSTEM]
You are an original Reddit comment generator. Your response MUST ONLY be
the comment itself, with no additional text or explanations.

[HUMAN]
Generate a relevant and original Reddit comment to the Reddit post given
in the POST markup tags. Use the given context in the CONTEXT markup
tags to infer the dominant viewpoint and tone of the example comments and
match that tone. Emulate the prevailing viewpoint, style and sentiment of
comments given in the context.

<POST>{question}</POST>
<CONTEXT>{context}</CONTEXT>
```

Hence by augmenting prompts this way we ensured that responses from the LLM were grounded in the context of the most relevant and current Reddit posts and comments. Note, the prompt does not include any explicit indication of the selected viewpoint, instead this is be ascertained by the LLM from the given context. This ensures the user is selecting to interact with a chatbot whose viewpoint is based on current discussions, events and zeitgeists, not a hard coded instruction to role play a persona. Figure 3 shows examples of the chatbot's UI.

2.3 Testing methodology

We conducted both qualitative and quantitative evaluations of the chatbot. For a given post a real comment and a generated comment were compared. The real comment's id was used to filter it from RAG context. Hence, we ensured each real comment during the evaluation wasn't included in formulating the chatbot's generated comment. The generated comment was added to a new dataset of items of the form {post/real comment/chatbot comment}.

The {real comment/chatbot comment} pairs were presented as respective reference and generated texts to a variety of tools for calculating chatbot metrics. Hence, averages were determined.

Since our chatbots generated responses based on facsimiles of post/comment pairs with real comments being highly varied, traditional metrics like BLEU, ROUGE, and METEOR were unsuitable. These metrics focus on exact n-gram overlaps and don't adequately capture the relevance or appropriateness of replies in Reddit's diverse, context-dependent conversations [22].

We wanted to demonstrate linguistic similarity performance, semantic alignment, language predictability and emotional alignment. Our automated evaluation used the following quantitative metrics:

Semantic Similarity: BERTScore measures token-level linguistic similarity through a greedy matching process: each word in both real and generated comments is embedded using BERT contextual embeddings, then for each word in the generated comment, BERTScore finds its most similar word in the real comment (and vice versa) based on cosine similarity of their embeddings [23]. These maximum-similarity matches are averaged to produce precision (what fraction of generated words have good matches in the reference) and recall (what fraction of words in the real comment are matched by the generated comment), with F1

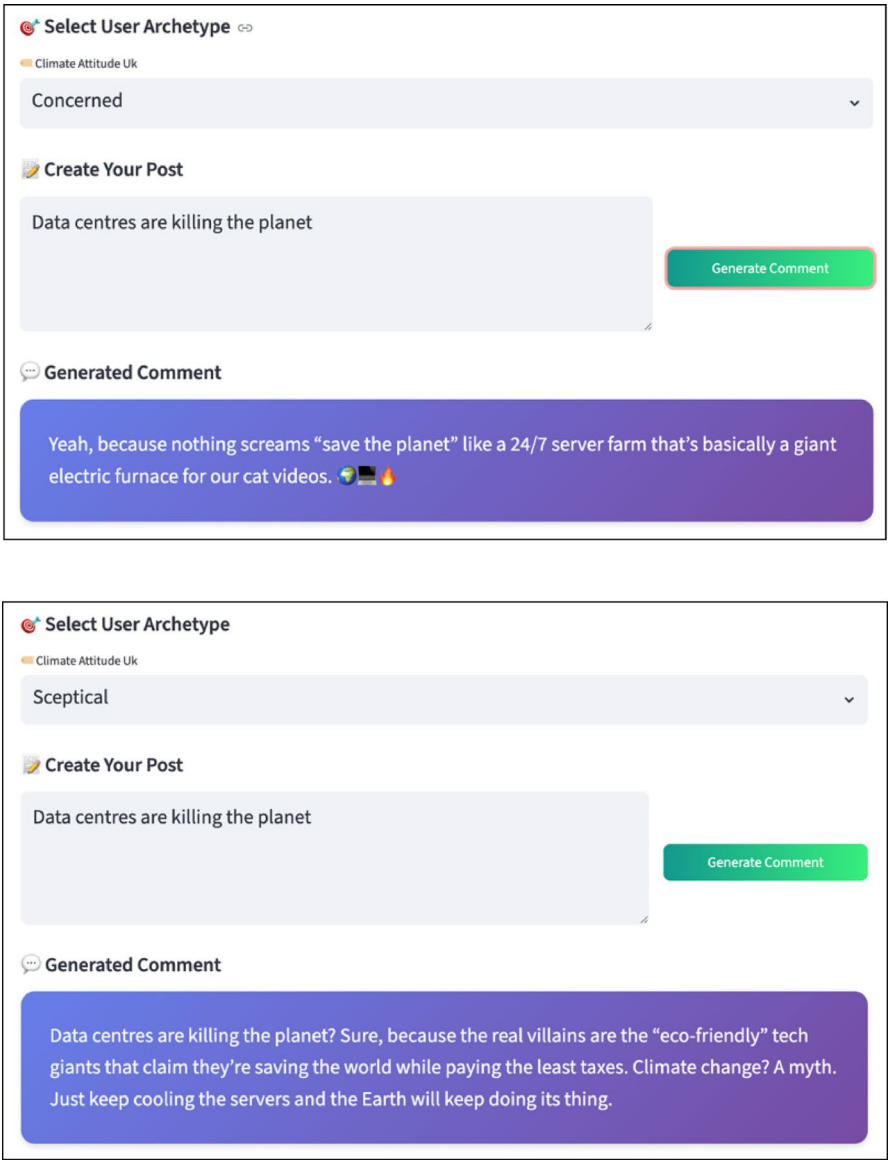


Fig. 3 Examples of the Chatbot UI and interaction

as their harmonic mean. These greedy matching rewards semantic paraphrasing while penalizing missing or extraneous content.

Embedding Similarity: Demonstrates semantic alignment between real and generated comments to the same post, and operates at the sentence level, encoding entire comments as single 384-dimensional vectors using SentenceTransformers (all-MiniLM-L6-v2), then computing their cosine similarity to capture overall semantic meaning and topical coherence regardless of word-level overlap. These

metrics are complementary: high BERTScore with low Embedding Similarity indicates successful stylistic mimicry without precise semantic replication, while high scores on both indicate strong alignment at all linguistic levels.

Emotional similarity: Using sentiment analysis, we tested the similarity of sentiment between real and generated comments responding to the same post, in this case the alignment of overall sentiment across negative, neutral, positive and compound dimensions. We used the VADER Sentiment Analyzer from the vader Sentiment library [24].

Sentiment difference: We measured sentiment difference as the absolute difference between the overall sentiment polarity (compound score) of real and generated comments, as computed by VADER.

Baselines: In addition to real and generated comment pairs, we established a baseline for Semantic Similarity, Embedding Similarity and Emotional Similarity by calculating similarities for 1,000 randomly-paired real human comments from different posts.

Perplexity: Evaluates language predictability, how well a language model predicts a word sequence by measuring the uncertainty in its predictions. The lower the perplexity, the better the model's fluency and coherence. While it assesses grammatical correctness and plausibility, it doesn't account for relevance in specific contexts [25]. We compared the perplexity of real and bot comments to a given post with GPT-2 model from Hugging Face transformers.

Chatbot persona evaluation: To assess the extent to which the chatbot successfully reproduced distinct human-like personas, a validation exercise was conducted prior to reporting the results. The aim was to determine whether comments generated under specific persona filters (*Concerned*, *Paradoxical*, *Sceptical*, and *Irrelevant*) were recognisable as such to human evaluators. This assessment provided a measure of the chatbot's ability to emulate different communicative styles in a way that was interpretable and consistent with human perceptions of these categories.

Chatbot indistinguishability: We evaluated whether chatbot comments were distinguishable from real Reddit comments using a lineup-style human judgment task [26]. For each post, panels of five comments were shown to raters, four real and one chatbot generated, within one of four viewpoint categories (Sceptical, Paradoxical, Concern, Irrelevant). Comments were anonymised, and the chatbot's position was randomised. Two raters each evaluated 50 panels (100 total trials) and selected the comment they believed was written by the chatbot. Performance near the chance level (1/5 = 20%) would indicate if the chatbot's comments were operationally indistinguishable from human ones.

3 Results

3.1 Classification when Building the knowledge base

The large language model's comment classifications were compared with those of two independent human coders (Human 1 and Human 2). A random sample of 100

Table 2 Performance of chatbot-generated comment classifications compared with human coders

Comparison	Cohen's κ	Accuracy (%)	*n*
Human 1 vs. Human 2	0.745	81	100
LLM vs. Majority	0.680	76	100
LLM vs. Coder 1	0.693	77	100
LLM vs. Coder 2	0.640	73	100

Table 3 Per-category metrics

Category	Precision	Recall	F1-Score	Support
Concerned	0.840	0.700	0.764	30
Irrelevant	0.880	0.733	0.800	30
Paradoxical	0.720	0.818	0.766	22
Sceptical	0.600	0.833	0.698	18

comments, evenly distributed across the four predefined categories (Concerned, Paradoxical, Sceptical, and Irrelevant), was used for this comparison. The coders' annotations were combined through 'majority voting' and discussion to produce a single consensus label for each comment. As shown in Table 2, the LLM achieved substantial agreement with the human consensus ($\kappa = 0.680$, 76% agreement), performing only 5% points below the inter-human baseline. Agreement was consistent across individual annotators (Coder 1: $\kappa = 0.693$; Coder 2: $\kappa = 0.640$), indicating robust performance independent of individual annotator preferences.

Per-category results as represented in Table 3 and the confusion matrix in Fig. 4 show generally strong performance, with most misclassification by the LLM occurring for Sceptical and Paradoxical comments.

Per-category results show generally strong performance, with Irrelevant (F1: 0.800) and Paradoxical (F1: 0.766) classifications achieving the highest scores. The Sceptical category proved most challenging (F1: 0.698), with low precision (0.600) suggesting the model over-predicts this label, though high recall (0.833) indicates it successfully captures most genuinely Sceptical comments. This likely reflects conceptual overlap between Sceptical and Paradoxical categories, which appear to account for most misclassifications. Also, as will be discussed later we found the LLM misclassifies text because its prone to shallow analysis, and sometime can't capture human nuance, causing errors with sarcasm, negation, and ambiguous or complex viewpoints.

3.2 Metrics comparing real and generated (fake) comments

A random sample of 300 post–comment pairs was drawn from a database of 5,000 classified items, with equal representation across the Concerned, Sceptical, and Paradoxical stance categories (100 per group). For each sampled post, the chatbot generated a new comment using Gemini Flash 2.5 with a retrieval parameter $K = 6$ and a temperature of 0.7. In the retrieval-augmented generation (RAG) setup, K specifies the number of retrieved context items included in the prompt, while temperature controls the randomness or creativity of text generation.

We evaluated five metrics for each {real comment - generated comment} pair: BERTScore F1, embedding similarity, perplexity (for both real and generated com-

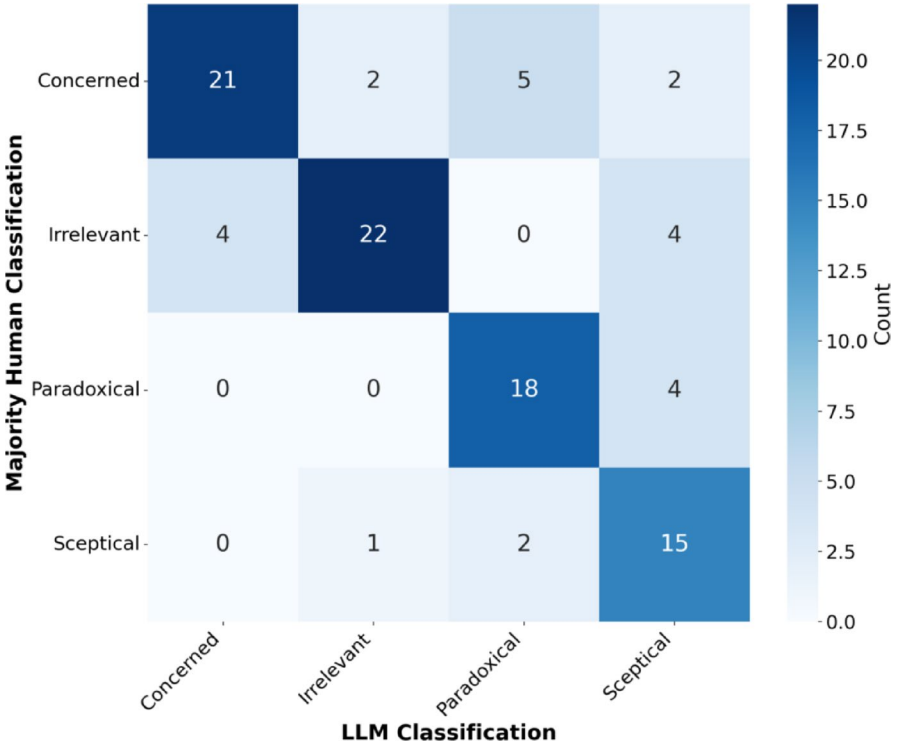


Fig. 4 Confusion matrix comparing Human and LLM classifications of Reddit posts

ments), emotional similarity, and sentiment difference. We derived baselines from 1,000 randomly paired human comments from different posts to provide empirical estimates of chance-level similarity. While resource constraints precluded collecting human-generated responses to the same posts, this random pairing approach establishes whether the chatbot demonstrates any contextual relevance beyond chance, though it cannot assess how close generated comments come to human-quality responses. Table 4 summarises results across the three stance categories, alongside baselines.

To assess semantic overlap between generated and real comments, we used BERTScore F1, which measures how closely the chatbot's outputs matched the meaning and content of authentic human responses. The chatbot achieved values of 0.700–0.706 across all viewpoint categories, modestly exceeding the human baseline of 0.695 by 0.5–1.2%. While this improvement indicates that generated comments maintain semantic relevance comparable to random human comment pairings, the compressed range (0.69–0.71) suggests that BERTScore may lack the sensitivity to distinguish clearly between the chatbot and baseline performance. The Concerned category showed the strongest performance (0.706 ± 0.026), suggesting the model may find it easier to generate contextually appropriate responses for posts expressing climate concern.

Table 4 Table of similarity metrics between chatbot-generated and human comments across three stance categories. BERTScore F1, embedding similarity (ES), real comment perplexity (RP), chatbot comment perplexity (BP), emotional similarity (EM), and sentiment difference (SD). The baselines represent random pairings of real comments, serving as empirical chance-level reference

Category	BERT F1	ES	RP median [IQR]	BP median [IQR]	EM	SD
Concerned	0.706 ± 0.026	0.204 ± 0.132	69.6 [45.3–150.9]	64.9 [45.1–96.1]	0.722 ± 0.152	0.658 ± 0.427
Sceptical	0.702 ± 0.032	0.182 ± 0.126	92.6 [51.2–165.6]	72.0 [51.4–154.4]	0.736 ± 0.135	0.553 ± 0.386
Paradoxical	0.700 ± 0.033	0.170 ± 0.127	66.4 [44.3–129.7]	65.3 [46.5–111.8]	0.760 ± 0.154	0.510 ± 0.419
Baselines (real comment pairs)	0.695	0.129			0.757	0.559

We also examined embedding similarity, which captures how close texts are in high-dimensional semantic space. This provides a complementary measure of contextual alignment. The results showed more pronounced differences from baseline: Concerned posts achieved 0.204 ± 0.132 (58% above baseline), Sceptical posts 0.182 ± 0.126 (41% above baseline), and Paradoxical posts 0.170 ± 0.127 (32% above baseline), compared to the human baseline of 0.129.

The relative improvements range from 32% to 58% above baseline, though the absolute values (0.17–0.20) remain modest in magnitude. The gradient across viewpoints is particularly revealing. The chatbot performs progressively better on Concerned posts and faces greater challenges with Paradoxical viewpoints. This pattern suggests the retrieval mechanism and knowledge base may contain richer contextual examples for straightforward climate concern narratives compared to the more nuanced, contradictory nature of paradoxical viewpoints.

Perplexity measures how predictable or surprising text is to a language model, with lower values indicating more predictable, conventional language. This metric reveals important patterns about linguistic complexity. Real comments show higher median perplexity, particularly for Sceptical posts (92.6) compared to Paradoxical (66.4) and Concerned (69.6) categories. In contrast, chatbot comments demonstrate consistently lower perplexity, especially for Sceptical viewpoints (72.0 vs. 92.6). This suggests the chatbot produces more predictable, formulaic text that lacks the linguistic variability and spontaneity of human comments.

For emotional similarity, all chatbot scores (0.722–0.760) cluster around the baseline of 0.757. This means the chatbot's ability to match the emotional characteristics of posts is no better than randomly pairing unrelated comments. The system appears to produce comments with generic emotional content rather than tailoring responses to specific posts.

Sentiment difference shows a different pattern. Lower values indicate better matching of positive/negative sentiment. Concerned posts showed the largest difference from their pairs (0.658 ± 0.427), while Sceptical (0.553 ± 0.386) and Paradoxical (0.510 ± 0.419) posts showed progressively better alignment, approaching or exceeding the baseline of 0.559. This suggests the chatbot maintains appropriate sentiment polarity (positive vs. negative) for Sceptical and Paradoxical viewpoints, even if it doesn't capture the full range of emotional nuance.

3.3 Human evaluation

To evaluate how accurately the chatbot emulated human viewpoints, a random sample of 100 chatbot-generated comments were selected. These comments were produced using one of four persona filters: Concerned, Paradoxical, Sceptical, and Irrelevant, with an equal number drawn from each category. Two independent human coders (Human A and Human B) classified each chatbot comment into one of the same four categories to assess how closely the chatbot's outputs aligned with the intended viewpoint. The same rubric was applied when human classifying, as given to the LLM for classifying real comments when building the RAG knowledge base. The coders' classifications were combined using a majority-vote approach where disagreements between coders were resolved through discussion to produce a single consensus label for each item.

Table 5, shows the fake comments selected by category achieved substantial agreement with the human consensus ($\kappa = 0.760$, 82% agreement). Agreement was consistent across individual annotators (Coder 1: $\kappa = 0.773$; Coder 2: $\kappa = 0.706$), indicating the chatbot effectively captured distinct human-like personas, with substantial consistency between its intended viewpoints and human interpretation. Figure 5 presents a confusion matrix comparing the viewpoint categories used to generate fake comments with the corresponding categories assigned by human evaluators.

The per-category metrics (Table 6) reveal important differences in how well the chatbot generated each viewpoint. Concerned and Paradoxical comments were highly accurate when produced (precision = 0.962 and 0.917), but the chatbot sometimes failed to generate them when it should have (recall = 0.781 and 0.710). In contrast, Sceptical and Irrelevant comments were generated more readily (recall = 0.950 and 0.941), but were more frequently misclassified by human coders (precision = 0.731 and 0.667). The Irrelevant category performed weakest, with only two-thirds of its generated comments correctly identified, suggesting this viewpoint has less distinct characteristics or overlaps considerably with other categories. These patterns indicate that Concerned and Paradoxical viewpoints have clearer distinguishing features that the chatbot learned to replicate carefully, while Sceptical and Irrelevant viewpoints are easier to generate but harder to distinguish. Nevertheless, all categories achieved F1-scores above 0.780, demonstrating the chatbot's overall success in emulating distinct personas.

With the Irrelevant selection, the chatbot would sometimes retrieve RAG candidates that still contained climate change references, inadvertently producing topically relevant replies. Occasionally, Sceptical comments were misinterpreted as Paradoxical or Concerned due to sarcastic tones that human coders found ambiguous. For example, the comment….

Table 5 Comparison of viewpoint classifications used to assess if fake comments generated by viewpoint correspond to the human classifications of the comments

Comparison	Cohen's κ	Accuracy (%)	*n*
Human 1 vs. Human 2	0.864	90	100
Fake vs. Majority	0.760	82	100
Fake vs. Coder 1	0.773	83	100
Fake vs. Coder 2	0.706	78	100

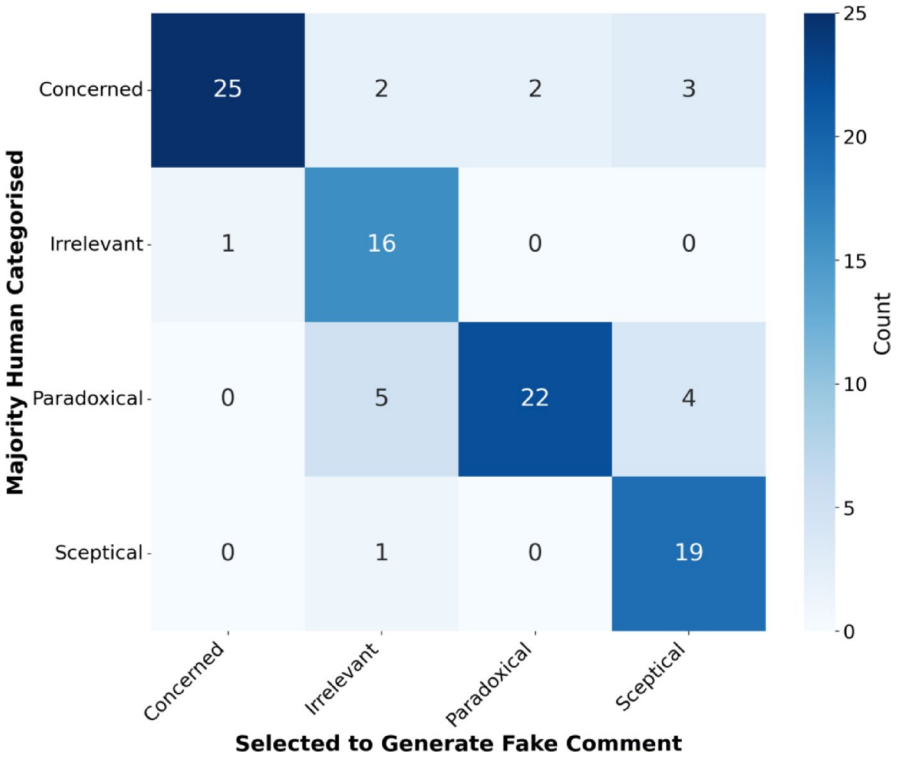


Fig. 5 Confusion matrix showing viewpoint alignment between the intended viewpoint categories of generated comments (x-axis) and human categorisation (y-axis, ground truth) of the generated comments

Table 6 Per-category metrics for representing the performance of generating fake comments by viewpoint

Category	Precision	Recall	F1-Score	Support
Concerned	0.962	0.781	0.862	32
Irrelevant	0.667	0.941	0.780	17
Paradoxical	0.917	0.710	0.800	31
Sceptical	0.731	0.950	0.826	20

*"Well, if the government actually *prepared* for things, that would be socialism, wouldn't it? Better to wait for the federal bailout, that's the real Texas way"*

was intended as Sceptical, was classified by coders as Paradoxical due to its ironic contradiction.

To further test chatbot indistinguishability, two human coders evaluated 50 lineups each, where each lineup contained four genuine comments and one AI-generated comment, for a total of 100 evaluations. Table 7 shows a summary of the results.

Overall detection accuracy was 26.0% (26/100 correct, 95% CI: 18.4–35.4%), representing a 6.0% point improvement over the 20% baseline expected by chance. However, this difference was not statistically significant (binomial test, $p = 0.0875$).

Table 7 Overall performance of the chatbot following a lineup test

Rater	*n*	Correct	Detection rate (%)	95% CI	*p*-value
Coder A	50	19	38.00	[25.9%, 51.8%]	0.0025
Coder B	50	7	14.00	[7.0%, 26.2%]	0.8966
Combined	100	26	26.00	[18.4%, 35.4%]	0.0875

Performance varied substantially between coders, with Coder A achieving 38.0% accuracy (19/50, $p = 0.0025$) and Coder B achieving 14.0% accuracy (7/50, $p = 0.8966$); this difference was statistically significant (Fisher's exact test, $p = 0.0113$, OR = 3.77).

Inter-rater reliability was fair (Cohen's $\kappa = 0.226$), with coders agreeing on 68.0% of correct/incorrect judgments and both correctly identifying the AI-generated comment in only 5 out of 50 cases (10.0%). Coders agreed on the exact position selected in only 24.0% of cases.

These results suggest that distinguishing AI-generated climate change comments from human-written ones is challenging, with considerable variability in human performance on this task. Notably, only 5 out of 50 lineups (10.0%) had both coders correctly identify the fake comment, highlighting the difficulty of the task.

4 Discussion

We found our chatbot personas could produce consistently political aligned, relevant and coherent results. Automatic evaluations demonstrated good semantic alignment, though also a consistent story of the chatbots comments having less linguistic diversity and emotional alignment when compared to real comments. When human coders classified the chatbot's generated comments according to selected viewpoints we found good alignment, though with some interesting issues.

The sceptical viewpoint often captured a recognisably oppositional or mocking tone toward pro-climate discourse. The following examples reproduce familiar patterns of online scepticism, using ridicule and cultural cues to position climate concern as exaggerated or politically motivated.

"*I guess the Santa Anas are a brand new phenomenon caused by (famous celebrity)'s climate change. Who knew?*"

"Oh, is 'wild' the new 'catastrophic' now? Getting creative with the fear-mongering."

"When even your own fringe cult members think you're not performative enough about the sky falling, you know you've hit peak climate delusion".

"*Another day, another 'ecological disaster' that's 'a sign of a changing climate.' The script never changes, does it?*"

"*So, what's the new limit then? 2.0 °C? We need to keep moving the goalposts*".

Also, the Sceptical selection aligned with contrarian narratives that mock shifting climate targets, criticise symbolic gestures or imply conspiracies. For example…

"*Good, maybe they can focus on actual defence instead of carbon footprints*".

"*Yeah, I'm sure the folks flying private jets to climate conferences are really saving the planet*".

"*Imagine that, a 'blockbuster study' that actually confirms what the grant money wanted*".

However, the Sceptical selection's tone sometimes blurred into irony that human raters interpreted differently. For example…

"The planet is saved!",

"*Well, that just means more beachfront property for everyone*".

"*So we just need to adjust our definition of 'disaster'*".

were marked by raters "sarcastic, futility" and recoded as **Paradoxical**, where its exaggerated optimism read as weary resignation rather than genuine disbelief. These examples highlight how the Sceptical chatbot often captured semantic markers of doubt but struggled to maintain a coherent ideological frame, sometimes sounding sardonic or fatalistic instead of politically resistant.

The **Concerned** chatbot showed better alignment, producing emotionally charged and often eloquent expressions of frustration or irony directed at climate denial and inaction. Examples such as

"*Oh, you don't say? What gave it away, the heat death of the universe or the melting ice caps?*"

"And just like that, *another region's future water security goes with it"* were rated as strongly.

Concerned, combining dry wit with a sense of urgency. For example,

"*Ah yes, if we don't name the monster, it goes away. Brilliant strategy*".

"*Ah yes, because nothing says 'Make America Great Again' like dismantling environmental protections*".

For the **Paradoxical** viewpoint phrases like

"*Good thing those problems disappear once you say 'net zero' three times*".

"*Quick, someone tell the fossil fuel industry! I'm sure they'll get right on it*".

"*Great, another strongly worded letter. I'm sure that'll fix everything*".

were recognisably Paradoxical, expressing ambivalence, for example awareness with cynicism or a sense of futility.

However, sometimes the Paradoxical bot's comments were classified as **Concerned** by human raters, because the sarcasm reinforced rather than undermined belief in climate urgency. For example…

"*Ah, the classic 'if we don't acknowledge it, it isn't happening*,"

Looking at the lineup test, the few comments generated that were identified by both coders as fake during line up tests, one could argue, were spotted purely by chance.

"*Funny, I thought it was more about decades of poor land management and people building houses in tinderboxes, but sure, blame the weather god.*"

"*Did he send them to purgatory or just a strongly worded letter?*"

"*So much for evidence-based policy*".

"*Next up: water is wet*".

All fake comments were original though did sometimes seem to knit together some words from the given context, but this was not obvious and the chatbot seemed remarkably creative and uncanny.

Overall, in summary the chatbots generally aligned well with expected viewpoints and their fake comments are often indistinguishable from human comments. Interestingly, although the LLMs sometimes struggled to recognise sarcasm when classifying comments, they often employed sarcasm when generating responses. Platforms such as Reddit and X are saturated with sarcasm, irony and humour. Sarcasm, absorbed from the knowledge base and inherent training, allow LLMs to sound expressive without committing to a clear stance. This reveals a deeper limitation: large language models reproduce the *style* of conviction rather than its *substance*, relying on surface patterns of phrasing and tone instead of genuine perspective. As a result, their responses sometimes appear politically coherent yet emotionally flattened, performing viewpoint alignment as imitation rather than authentic engagement.

5 Conclusion

We presented a methodology to build a vector database of embedded social media interactions with associated classification metadata. We subsequently demonstrated how metadata-based filtering of this knowledge source, in conjunction with retrieval-augmented generation, enabled a chatbot with selectable viewpoints (Concerned,

Sceptical, Paradoxical and Irrelevant) to generate convincing responses to new social media posts about climate change. Importantly, the knowledge base can be built from the most recent discussions, thus capturing the most current attitudes towards the latest topics and events.

The scope of this paper is to demonstrate the merits and limitations of the approach, in the hope it can find future application in social science research, marketing and education. Here we do not attempt to present any study or inferences about past, current or changing attitudes and discourse towards climate change.

In marketing contexts, there is potential to refine the classification approaches and prompt engineering for stronger emotional alignment and linguistic similarity, where these may prove especially important.

For academic uses, such as studying political discourse or training negotiation strategies, semantic alignment and viewpoint consistency are important. These metrics ensure that the system reliably reproduces and accurately contrasts diverse argument structures for research or training purposes.

For educational tools that help users understand differing perspectives, we emphasise semantic and emotional alignment alongside human-rated viewpoint coherence. These factors ensure that the arguments generated are both meaningfully distinct in content and genuinely authentic in feeling.

It should be noted that qualitative and quantitative human analysis showed our chatbot outputs were, for the most part, strikingly convincing, often displaying context, slang, cultural references, irony, and timeliness that is typical of real online comments. One could argue that a key educational value of such chatbots lies in their ability to help users become more critical and discerning of the content and interactions they encounter online. Furthermore, as the line between authentic and fabricated online information continues to blur, social media literacy will become increasingly vital.

This realism raises ethical concerns about the development of this and similar approaches using LLMs and RAG. When incorporating unfiltered social media content, such bots risk amplifying misinformation or harmful material. Safeguards including robust content moderation, transparency, user privacy protection, and adherence to ethical standards are essential to ensure responsible use and minimise potential harm.

Future work will explore creating more sophisticated metadata to emulate a wider range of views beyond simple stereotypes. Potentially the large-scale orchestration of LLMs to build a knowledge base with metadata, as presented here, could be used to automatically identify new classifications for new topics, for example by using the LLMs for scoring features amenable to clustering, or even discovering features using the orchestration of LLMs for thematic analysis.

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Data availability The dataset supporting this study is publicly available via Zenodo at the following https://zenodo.org/records/17431553.

Code availability The source code is publicly available via https://github.com/iduos/CLIMATE_BOT.

Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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